TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Low Resistance by the Fall of Potential Method

For resistances of a fraction of an ohm, the leads themselves are the error. Let one current flow through unknown and standard in series, and compare only the voltages that fall across each — the leads never enter the comparison.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 10 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to measure a low resistance (of order 0.1 Ω) by comparing falls of potential. The unknown X and a standard low resistance S are connected in series so that *the same current I* flows through both. A sensitive galvanometer (used as a voltmeter, with a protective series resistance) is connected first across S, then across X, through separate **potential leads** attached directly at the ends of each resistance. The deflections are proportional to the potential differences:

$$\theta_S \propto V_S = I\,S \quad \cdot \quad \theta_X \propto V_X = I\,X \quad \rightarrow \quad X = S \times \theta_X/\theta_S$$

The current cancels — it need not even be known, only steady. And because the potential leads carry only the galvanometer's tiny current, the resistance of the main current leads and their contacts (which may rival X itself) never appears: this is the **four-terminal** principle, the only honest way to measure small resistances.

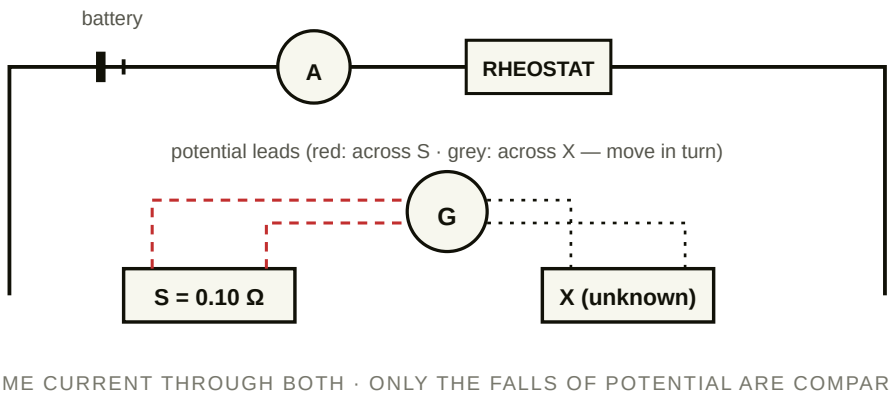


Figure 1 — Fall of potential: the heavy loop carries the working current through S and X in series; the galvanometer visits each resistance in turn through its own light potential leads, which attach inside the current terminals.

2. Apparatus and procedure

- **Apparatus:** standard low resistance (0.1 Ω, four-terminal), the unknown low resistance (a short thick wire, an ammeter shunt, a coil), battery, rheostat, ammeter, sensitive galvanometer with protective series resistance, reversing key, thick connecting leads for the current loop.
- Wire the current loop: battery → rheostat → ammeter → S → X → back to battery, all with short thick leads and firm terminals.
- Attach the potential leads across S (at S's own potential terminals). Set the rheostat for a steady current of a couple of amperes. Read the galvanometer deflection θ_S; reverse the current and average.
- Without disturbing the current, move the potential leads across X and read θ_X the same way. Alternate S, X, S, X a few times to expose any drift of the current, and average each.
- Compute $X = S \cdot \theta_X / \theta_S$. Repeat at a second current as a check — the ratio should not change.

3. Worked calculation

Standard S = 0.10 Ω. Mean deflections: across S, θ_S = 40 divisions; across X, θ_X = 53 divisions (same steady current, about 2 A).

$$X = 0.10 \times 53/40 = 0.1325\ \Omega \approx 0.133\ \Omega$$

(In volts: $V_S = 2 \times 0.10 = 0.20$ V and $V_X = 2 \times 0.1325 = 0.265$ V — the deflections stand in exactly that ratio, 53 : 40.)

4. Precautions

- Potential leads must touch the resistance *inside* its current terminals — clip them onto the body of S and X themselves, never onto the current leads, or the comparison quietly includes a contact resistance.
- The current must not drift between the S and X readings; alternate the readings and watch the ammeter.
- Keep the current's duration short: 2 A through 0.1 Ω warms both resistances, and their values (and the battery) drift.
- Reverse the current and average, to cancel thermal emfs at the dissimilar-metal junctions, which at millivolt levels are real voltages, not noise.

Practice (viva preparation)

1. Why does the ordinary two-terminal ohmmeter arrangement fail at 0.1 Ω?
2. Deflections: across a 0.05 Ω standard, 32 div; across the unknown, 45 div. Find X.
3. Why need the current be only steady, not known?
4. What are thermal emfs and why does current reversal cancel them?
5. The potential leads are clipped onto the current leads just outside X's terminals. What error enters?

Answer key

1. Lead and contact resistances (easily 0.01–0.1 Ω in total) sit in series with the unknown and are indistinguishable from it — the error is the same size as the quantity. The four-terminal comparison removes them because the potential leads carry essentially no current and so drop no voltage.
2. X = 0.05 × 45/32 ≈ 0.070 Ω.
3. Both falls of potential are proportional to the same I; in the ratio θ_X/θ_S = IX/IS the current divides out. Only a drift *between* the two readings corrupts the ratio — hence steady, not known.
4. At junctions of dissimilar metals, temperature differences generate small emfs (μV–mV). They add to the measured fall in one current direction and subtract in the other, so averaging direct and reversed readings cancels them while the true IR fall (which reverses sign with I and is read by magnitude) survives.
5. The measured "fall across X" then includes the IR drop of the short lead pieces and contacts between the clip points and X's true ends — an extra resistance added to X, exactly the error the method exists to avoid.

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